Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2000-II-Q04] | Part A (Q1-Q5)

Answer: 180

1. Write $N = 2^a m$ with m odd. Odd divisors number $\tau(m) = 6$, while even divisors number $a\tau(m) = 12$, so $a = 2$.

2. The odd factor with six divisors has form p^5 or $p^2 q$ for distinct odd primes. Their minima are 243 and 45, respectively. Hence the least N is $4 \cdot 45 = 180$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-I-Q02] | Part A (Q1-Q5)

Answer: 012

1. The condition is exactly $k \mid 2004$, since the target requires $1 + jk = 2005$ for an integer $j \geq 1$.

2. $2004 = 2^2 \cdot 3 \cdot 167$, so the divisor count is $3 \cdot 2 \cdot 2 = 12$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-I-Q03] | Part A (Q1-Q5)

Answer: 109

1. Four total divisors forces the form p^3 or pq for distinct primes. The latter has proper divisors $1, p, q$, so choose two of the fifteen primes below fifty: $\binom{15}{2} = 105$.

2. For p^3, the largest proper divisor is $p^2 < 50$, giving $p = 2, 3, 5, 7$. Total $105 + 4 = 109$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-II-Q04] | Part A (Q1-Q5)

Answer: 435

1. The factorizations give divisor counts $11^2 = 121$, $8^2 = 64$, and $12 \cdot 23 = 276$. Pairwise intersections are divisors of gcds $5^7, 2^{10}, 3^7$, with counts $8, 11, 8$.

2. The triple intersection contains only one. Inclusion–exclusion gives $121 + 64 + 276 - 8 - 11 - 8 + 1 = 435$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2014-I-Q03] | Part A (Q1-Q5)

Answer: 200

1. Write the fraction $a/(1000-a)$ with $1 \leq a < 500$. Reduction is equivalent to $\gcd(a, 1000) = 1$.

2. There are $\varphi(1000) = 1000(1-1/2)(1-1/5) = 400$ coprime integers from one to 999. Pairing a with $1000-a$ puts exactly half below 500, with no fixed coprime point.

3. The answer is 200.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2004-I-Q08] | Part B (Q6-Q10)

Answer: 199

1. Equal turning and equal lengths put the vertices on a circle with a constant step d among the regular n-gon vertices. Visiting all vertices once requires $\gcd(d, n) = 1$. Crossing excludes the ordinary polygon step 1.

2. Selecting the counterclockwise orientation gives $1 < d < n/2$; equivalently pair the coprime steps $d, n - d$ and remove the pair $1, n - 1$.

3. $\varphi(1000) = 1000(1 - 1/2)(1 - 1/5) = 400$. Therefore the count is $400/2 - 1 = 199$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2004-II-Q08] | Part B (Q6-Q10)

Answer: 054

1. Factor $2004 = 2^2 \cdot 3 \cdot 167$. A candidate is $2^i 3^j 167^k$, whose divisor count is $(i+1)(j+1)(k+1)$. We require this product to equal 2004.

2. Each factor is at most 2004, so all resulting exponents fit the original power bounds. Distribute the two factors of 2 among three slots in $\binom{4}{2} = 6$ ways; each of $3, 167$ chooses one of three slots.

3. The count is $6 \cdot 3 \cdot 3 = 54$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2019-I-Q09] | Part B (Q6-Q10)

Answer: 540

1. Except $n = 1$, both divisor counts are at least two. Thus the pair must be $(2, 5)$ or $(3, 4)$ in either order. One number is therefore a prime square or prime fourth power.

2. Test neighbors of these powers in increasing order. Through the sixth successful value, the candidates are prime squares $4, 9, 25, 49, 121, 169, 289, 361$ and fourth powers $16, 81$. Only $n = 8, 9, 16, 25, 121, 361$ qualify; the corresponding neighbors have counts four or are prime as required.

3. Their sum is $8 + 9 + 16 + 25 + 121 + 361 = 540$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2019-II-Q09] | Part B (Q6-Q10)

Answer: 472

1. Write $n = 2^a 5^b m$ with $a \geq 2, b \geq 1, \gcd(m, 10) = 1$. Divisor counts require $(a+1)(b+1)\tau(m) = 20$. The possible $a+1 \geq 3$ are $4, 5, 10, 20$.

2. The cases four and ten yield $5000, 2560$, too large; twenty is impossible. For $a+1 = 5$, either $b = 3, m = 1$, giving 2000, or $b = 1, m = p$ prime other than two,five. The bound gives $p \in \{3, 7, 11, 13, 17, 19, 23\}$.

3. Hence $S = 2000 + 80(93) = 9440$, and $S/20 = 472$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2026-II-Q09] | Part B (Q6-Q10)

Answer: 669

1. Expand each summand geometrically and regroup nonnegative terms: $S = \sum_{n\geq 1} d(n)10^{-n}$, since exponent n occurs once for each divisor of n.

2. After multiplying by 10^{100}, terms with $n \leq 100$ are integral. The remaining tail starts with $2/10 + 8/100 + 2/1000 + 8/10000 = .2828$; for later terms use $d(100+j) \leq 100 + j$ to bound the rest by $.002$. Hence the whole tail is less than one and causes no carry.

3. Only the last three integral terms matter modulo 1000: $100d(98) + 10d(99) + d(100) = 100 \cdot 6 + 10 \cdot 6 + 9 = 669$.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2005-I-Q12] | Part C (Q11-Q15)

Answer: 025

1. Divisors pair except at squares, so $\tau(j)$ is odd exactly for square j. Hence $S(n)$ has parity $\lfloor\sqrt{n}\rfloor$.

2. Odd blocks correspond to $k = 1, 3, \ldots, 43$, each containing $2k+1$ integers from k^2 through $(k+1)^2 - 1$. They are all complete below 2005. Thus $a = 2(1 + 3 + \cdots + 43) + 22 = 990$.

3. Then $b = 1015$, and the absolute difference is 25.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2006-II-Q13] | Part C (Q11-Q15)

Answer: 015

1. Such a representation is $N = m^2 - k^2 = (m-k)(m+k)$ with integers $m > k \geq 0$. Thus it corresponds to a factor pair $a \leq b$ of equal parity, with length $j = a$.

2. For odd N, this is five unrestricted divisor pairs, requiring divisor count nine or ten. Below one thousand, the odd possibilities are $225, 441, 405, 567, 891$, giving five.

3. For even N, both factors are even, so $N = 4M$ and $M < 250$ must have nine or ten divisors. The possibilities are $36, 100, 196, 225$ and $48, 80, 112, 176, 208, 162$, giving ten. Total 15.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2016-II-Q11] | Part C (Q11-Q15)

Answer: 749

1. Prime exponents e_i of a give divisor count $\prod(ke_i+1) \equiv 1 \pmod{k}$. Conversely every $N = ke+1$ occurs using $a = 2^e$, including $N = 1$ via $a = 1$. Thus k-nice means exactly $N \equiv 1 \pmod{k}$.

2. From1 through999, the counts congruent to1 modulo7 and8 are143 and125. Their intersection is1 modulo56, with18 values.

3. Inclusion-exclusion leaves $999 - (143 + 125 - 18) = 749$ integers of neither kind.

Source / 来源

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

[AIME-2021-I-Q14] | Part C (Q11-Q15)

Answer: 125

1. Factor $2021 = 43 \cdot 47$. For a prime power factor of a, its divisor sum is $1 + p + \cdots + p^{en}$. It suffices that the nonconstant part vanish modulo each $q = 43, 47$.

2. If $p \equiv 0 \pmod{q}$ it vanishes automatically; if $p \equiv 1$, it requires $q \mid en$; otherwise $p^{en} = 1$ suffices. Thus $q(q-1) \mid n$ is sufficient.

3. The q factors are necessary by taking prime $a = 173 \equiv 1 \pmod{43}$ and $a = 283 \equiv 1 \pmod{47}$. For $q - 1$, take primes $a = 3$ of order 42 modulo 43 and $a = 5$ of order 46 modulo 47.

4. Hence $n = \operatorname{lcm}(43 \cdot 42, 47 \cdot 46) = 2 \cdot 3 \cdot 7 \cdot 23 \cdot 43 \cdot 47$. Its prime-factor sum is 125.

Source / 来源